Proving the Rank-Nullity Theorem from Linear Independence

MH1200 Linear Algebra I

Hans Gabriel D. De Vera
Nanyang Technological University
hansgabr001@e.ntu.edu.sg

1. Introduction

This document proves the Rank-Nullity Theorem, starting from everything covered up until Linear Independence.

2. Basis

Definition Basis

Definition 1

A basis S of a subspace $T \subset \mathbb{R}^n$ is a **list** of linearly-independent vectors that spans the subspace T.

Theorem Minimality of a Basis

Theorem 2

Suppose a subspace T has a basis S with n vectors.

Further suppose that we have a list of m vectors S' that spans T. Then, $m \geq n$.

Proof Minimality of a Basis

Proof 3

Suppose, for the sake of contradiction, that there exists a list of m vectors S' that spans T and a basis S of T consisting of n vectors such that $m < n$.

Let $\vec{v_i} \in S$. Then, for each such $\vec{v_i}$, there exist $\vec{w_j} \in S'$ and $c_j \in \mathbb{R}$ such that:

$$\vec{v_i} = \sum_{j=1}^{m} c_j \vec{w_j}$$

TODO proof route:
- Show that the elementary row operations preserve the row space
- Show that a basis is linearly independent iff no zero rows when vectors are encoded as row vectors in a matrix
- Show that any non-minimal basis is necessarily linearly dependent
- Derive a contradiction from the above fact, conclude the result

TODO to show rank-nullity:
- Define the concept of dimension
- Show that the row spaces and column spaces always have matching dimensions
- Define rank as the dimension of the row space or column space
- prove stuff about rank and nullity w.r.t. the structure of a matrix
- prove the rank-nullity theorem